

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0790

Broadacre 345 kV
April 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0790. The Interconnection Customer proposes a 313.9 MW solar photovoltaic facility interconnecting at the Broadacre 345 kV substation in the Kentucky area. The request was received on April 30, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 244 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,021,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Beaver – Eufaua 500 kV
Contingency: Norwich – Omni-Hackberry Substation 500 kV
Loading: 109.2% (Rating: 3,064 MVA)
DFAX: 6.5%
Pre-project loading: 108.5%
2. Facility: Trego Substation – Swan Vly 500 kV
Contingency: Troy Energy Llc – Richland 500 kV
Loading: 100.6% (Rating: 3,663 MVA)
DFAX: 5.9%
Pre-project loading: 100.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beaver – Eufaua 500 kV	Reconductor line	\$23,079,000	3% (\$648,000)
Swan Vly – Trego 500 kV	Reconductor line	\$67,171,000	2% (\$1,373,000)
	Total allocated to project		\$2,021,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.